

CS425: Computer Networks

Homework-1 : 230316

Question 1

Cyclic Redundancy Check (CRC) is an error detection technique based on polynomial arithmetic over modulo-2 field. In CRC, binary sequences are interpreted as polynomials with coefficients in $\{0, 1\}$.

Let:

- $D(x)$ be the data polynomial
- $P(x)$ be the generator polynomial of degree r

To generate the CRC:

1. Multiply the data polynomial by x^r (append r zeros).
2. Divide $D(x)x^r$ by $P(x)$ using modulo-2 division.
3. Let the remainder be $R(x)$.
4. Transmitted frame is:

$$T(x) = D(x)x^r + R(x)$$

At the receiver:

$$T(x) \bmod P(x)$$

If remainder is zero $\rightarrow$ no error detected.

If remainder is non-zero $\rightarrow$ error detected.

Program Description

A C++ program was written to implement the CRC algorithm. The program:

1. Generates a 10-bit message.
2. Uses generator polynomial $P = 110101$.
3. Computes the CRC remainder.
4. Forms the transmitted frame.
5. Introduces a random single-bit error.
6. Applies CRC at the receiver.
7. Determines whether the frame is accepted or discarded.

Sample Output

```
Original Data: 1010101010
Generator Polynomial: 110101
CRC: 01000
Transmitted Frame: 1010101001000
Received Frame (error at position 14): 1010101001001
Frame Discarded (Error Detected)
```

The output confirms that a single-bit error produces a non-zero remainder, and the frame is correctly discarded.

How to Compile and Run

```
g++ crc.cpp -o crc
./crc
```

Question 2

In Go-Back-N (GBN) ARQ, suppose k bits are used for sequence numbering. Then the total number of distinct sequence numbers available is:

$$\text{Sequence number space} = 2^k$$

These sequence numbers are used modulo 2^k , meaning that after $2^k - 1$, numbering wraps back to 0.

Why cannot the window size be 2^k ?

Assume the sender window size is $W = 2^k$. Consider a concrete example with $k = 3$.

Then:

$$2^k = 8$$

Sequence numbers: 0, 1, 2, 3, 4, 5, 6, 7

Suppose the sender transmits all 8 frames:

$$0, 1, 2, 3, 4, 5, 6, 7$$

The receiver successfully receives all frames and sends a cumulative acknowledgment (e.g., RR 0 or RR 8 modulo 8).

Now assume this acknowledgment is lost.

Since the sender does not receive the ACK, it times out and retransmits frame 0.

However, at this point the receiver has already advanced its window and is expecting the *next cycle* of frames beginning again from 0.

Now when the receiver sees frame 0, it cannot distinguish between:

- A retransmission of the old frame 0 (due to lost ACK), and
- A new frame 0 from the next transmission cycle.

This ambiguity occurs because sequence numbers are reused modulo 2^k .

General Condition

To avoid ambiguity, the sender must not be allowed to send an entire sequence space worth of frames before receiving acknowledgments.

Thus, the window size must satisfy:

$$W < 2^k$$

Since window size must be an integer,

$$W_{\max} = 2^k - 1$$

This guarantees that at least one sequence number is always unused, which prevents overlap between old and new transmission cycles.

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Question 3

In Selective Reject (Selective Repeat) ARQ, both sender and receiver maintain sliding windows of size W .

Unlike Go-Back-N, the receiver in Selective Reject accepts and buffers out-of-order frames.

Sequence Number Space

With k -bit sequence numbers:

$$\text{Total distinct sequence numbers} = 2^k$$

Because numbering is modulo 2^k , ambiguity may arise if the sender and receiver windows overlap after wraparound.

Why is the condition stricter than GBN?

Suppose the sender window size is W .

The sender window spans W sequence numbers. The receiver window also spans W sequence numbers.

To avoid ambiguity between old and new frames, these two windows must not overlap after wraparound.

Thus, the total span of both windows together must not exceed the sequence number space:

$$W + W \leq 2^k$$

$$2W \leq 2^k$$

$$W \leq 2^{k-1}$$

Example: $k = 3$

Sequence space = 8.

If $W = 5$, then:

$$2W = 10 > 8$$

Overlap occurs $\rightarrow$ ambiguity.

If $W = 4$, then:

$$2W = 8$$

No overlap occurs.

Maximum Window Size

$$W_{\max} = 2^{k-1}$$

Thus, Selective Reject requires the window size to be at most half of the sequence number space.

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Question 4

Given:

- Data rate $R = 4 \text{ kbps} = 4000 \text{ bits/s}$
- Propagation delay $T_p = 20 \text{ ms} = 0.02 \text{ s}$

Stop-and-Wait Efficiency

Efficiency of Stop-and-Wait ARQ is defined as:

$$\eta = \frac{\text{Transmission time}}{\text{Transmission time} + 2 \times \text{Propagation delay}}$$

Let T_f be the transmission time of one frame.

$$\eta = \frac{T_f}{T_f + 2T_p}$$

We require:

$$\eta \geq 0.5$$

Solving the Inequality

$$\frac{T_f}{T_f + 2T_p} \geq \frac{1}{2}$$

Cross-multiplying (since denominator is positive):

$$2T_f \geq T_f + 2T_p$$

$$T_f \geq 2T_p$$

Substituting $T_p = 0.02 \text{ s}$:

$$T_f \geq 0.04 \text{ s}$$

Relating Transmission Time to Frame Size

Transmission time:

$$T_f = \frac{S}{R}$$

where S = frame size in bits.

$$\frac{S}{4000} \geq 0.04$$

$$S \geq 0.04 \times 4000$$

$$S \geq 160 \text{ bits}$$

$$\boxed{S \geq 160 \text{ bits}}$$

Hence, frame size must be at least 160 bits to achieve at least 50% efficiency.

Question 5

Given:

- Frame size = 4 bits
- Bit error probability $p = 10^{-3}$
- Errors are independent

(a) Probability of No Error

Each bit is correct with probability $(1 - p)$.

Since bits are independent:

$$P(\text{no error}) = (1 - p)^4$$

$$= (0.999)^4$$

Using binomial expansion:

$$(1 - p)^4 = 1 - 4p + 6p^2 - 4p^3 + p^4$$

Neglecting higher-order terms:

$$\approx 1 - 4(10^{-3})$$

$$\approx 0.996$$

$$\boxed{P(\text{no error}) \approx 0.996}$$

(b) Probability of At Least One Error

$$P(\text{at least one error}) = 1 - (1 - p)^4$$

$$\approx 1 - 0.996$$

$$= 0.004$$

$$\boxed{0.004}$$

(c) Undetected Error with One Parity Bit

After adding one parity bit:

Total bits = 5.

Parity detects all **odd** number of bit errors.

Undetected errors occur when an **even** number of bits flip.

Exactly 2 Errors

$$P(2) = \binom{5}{2} p^2 (1 - p)^3$$

$$= 10 \times (10^{-6}) \times (0.999)^3$$

$$\approx 9.97 \times 10^{-6}$$

Exactly 4 Errors

$$P(4) = \binom{5}{4} p^4 (1 - p)$$

$$\approx 5 \times 10^{-12}$$

Negligible compared to the 2-error case.

$$\boxed{P(\text{undetected}) \approx 9.97 \times 10^{-6}}$$

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Question 6

Given:

$$P = 110011, \quad M = 11100011$$

Polynomial Representation

Generator polynomial:

$$P(x) = x^5 + x^4 + x + 1$$

Message polynomial:

$$M(x) = x^7 + x^6 + x^5 + x$$

Since generator has degree 5, append 5 zeros:

$$M(x)x^5$$

Binary form:

$$11100011\ 00000$$

Modulo-2 Division

Divide:

$$1110001100000$$

by:

$$110011$$

Using XOR subtraction repeatedly, the remainder obtained is:

$$11010$$

CRC = 11010

The transmitted frame becomes:

$$11100011\ 11010$$

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Question 7

(a) Encoding

Given:

$$P(x) = x^4 + x + 1$$

Binary form:

$$10011$$

Message bits:

$$10010011011$$

Polynomial form:

$$M(x) = x^{10} + x^7 + x^4 + x^3 + x + 1$$

Multiply by x^4 :

$$x^{14} + x^{11} + x^8 + x^7 + x^5 + x^4$$

Divide by $P(x)$.

Remainder obtained:

$$R(x) = x^3 + x^2$$

Binary:

1100

Encoded frame:

100100110111100

(b) Error Pattern: 1000100000000000

Received frame:

$$T(x) + E(x)$$

Divide by $P(x)$.

Since remainder $\neq 0$, error is detected.

(c) Error Pattern: 1001100000000000

If the error polynomial $E(x)$ is divisible by $P(x)$, then:

$$P(x) \mid E(x)$$

Thus remainder becomes zero and error is not detected.

This demonstrates the fundamental CRC property:

An error is undetected iff $P(x)$ divides $E(x)$.

References

- Module 11, *Error Detection Techniques (Parity Check, Checksum, CRC)*
- Module 15, *Error Control and ARQ Protocols*